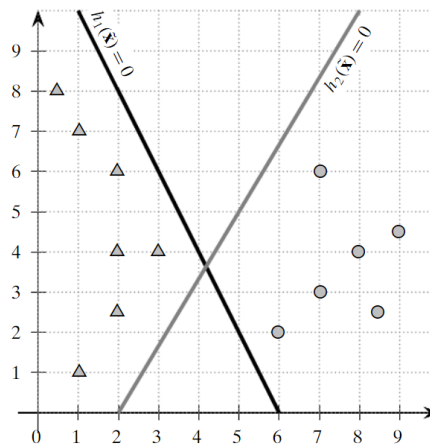


	Course Name:	Data Analysis and Visualization	Course Code:	DL3001
	Program:	BS Data Sciences	Semester:	Fall 2023
	Section	BS DS 5A	Total Marks:	10
	Date:	20th November 2023	Weight	
	Exam Type:	Quiz 3	Page(s):	2

Q1: Consider the following dataset, which has points from two classes c1(triangles)and c2(circles). Answer the questions below.



(a) Find the equations for the two hyperplanes h1 and h2.

[5 Marks]

For h1, we have the two points on the line, (6,0) and (1,10),
So, Slope $m_1 = \frac{10}{1-6}$
 $= -2$

Using (6,0), as the point, we get the equation of the line as
 $(x_2 - 0) / (x_1 - 6) = -2$
 $\Rightarrow 2x_1 + x_2 - 12 = 0$

Similarly, for h_2 , we have the two points on the line $(2,0)$ and $(8,10)$,
So, slope $m_2 = 10/(8-2)$
 $=5/3$

Using $(2,0)$, as the point, we get the equation of the line as

$$(x_2-0)/(x_1-2)=5/3$$
$$\Rightarrow 5x_1-3x_2-10=0$$

(b) Show all the support vectors for h_1 and h_2 .

[5 Marks]

The support vectors for h_1 are $(2,6)$, $(3,4)$ and $(6,2)$. The support vectors for h_2 are $(3,4)$ and $(7,6)$.